C# tips

SOLID Principles

SWIPE

S — Single Responsibility Principle (SRP)

A class should have one reason to change.

Bad

```
class InvoiceService
{
    public void CreateInvoice() { }
    public void SaveToDatabase() { }
    public void SendEmail() { }
}
```

Good

```
class InvoiceService
{
    public void CreateInvoice() { }
}

class InvoiceRepository
{
    public void Save() { }
}

class EmailService
{
    public void SendInvoice() { }
}
```

O — Open/Closed Principle (OCP)

Software entities should be open for extension, closed for modification.

Bad

```
class PaymentProcessor
{
    public void Pay(string type)
    {
        if (type == "Card") { }
        if (type == "Bank") { }
    }
}
```

Good

```
interface IPaymentMethod
{
    void Pay();
}

class CardPayment : IPaymentMethod
{
    public void Pay() { }
}

class PaymentProcessor
{
    public void Process(IPaymentMethod method) => method.Pay();
}
```

L — Liskov Substitution Principle (LSP)

Derived classes must be substitutable for their base classes.

Bad

```
class Bird
{
    public virtual void Fly() { }
}

class Penguin : Bird
{
    public override void Fly() => throw new NotImplementedException();
}
```

Good

```
interface IFlyingBird
{
    void Fly();
}

class Sparrow : IFlyingBird
{
    public void Fly() { }
}
```

I — Interface Segregation Principle (ISP)

Don't force classes to depend on methods they don't use.

Bad

```
interface IMachine
{
    void Print();
    void Scan();
    void Fax();
}
```

Good

```
interface IPrinter
{
    void Print();
}

interface IScanner
{
    void Scan();
}
```

D — Dependency Inversion Principle (DIP)

Depend on abstractions, not concrete implementations.

Bad

```
class NotificationService
{
    private EmailService _email = new EmailService();
}
```

Good

```
interface IMessageService
{
    void Send();
}

class EmailService : IMessageService
{
    public void Send() { }
}

class NotificationService
{
    private readonly IMessageService _service;

    public NotificationService(IMessageService service)
    {
        _service = service;
    }
}
```

Thanks for reading!

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